Roohana Karim

roohanak123@gmail.com | +1 469-348-8820 | GitHub | LinkedIn | Portfolio

EDUCATION

Louisiana State University | Baton Rouge, LA

Ph.D. in Computer Science | Aug 2023 - Present

M.S. in Computer Science | Aug 2023 - Dec 2025

SELECTED PUBLICATION

SoK: Anti-Forensics Evolution, Platform Coverage, and Mobile Residual Evidence

Coauthor | Accepted, DFRWS USA 2026

RESEARCH EXPERIENCE

Research Assistant - Bit Lab, Louisiana State University | Sep 2024 - Present

- Contributed to extracting, manually labeling, and verifying 7,865 dataset instances for a systematic study of datasets in computer security research.
- Built an LLM-based Python pipeline and developed prompts for related information-extraction and labeling tasks in a separate research project.

LLM Authorship and Source Attribution | Ongoing research

- Generated code artifacts from 10 locally hosted language models to investigate model-specific coding behavior and source attribution.
- Developing attribution baselines using character n-gram features and classifiers including XGBoost and random forests.
- Designing controlled ablation experiments varying temperature, top-p, and adversarial prompting to examine attribution robustness and model fingerprints.

SELECTED PROJECTS

Kitsune Network Attack Classification | GitHub

- Built a machine-learning workflow for malicious network activity detection with preprocessing, feature analysis, model training, and evaluation artifacts including ROC curves and feature importance.

SecureScholar - Cybersecurity Research Assistant | GitHub

- Built a RAG assistant using LlamaIndex, ChromaDB, Streamlit, and Ollama for semantic search, citation-backed retrieval, and cybersecurity literature analysis.

Savorly - AI Cooking Journal | React, TypeScript, Vite, Node.js | GitHub | Demo

- Built a recipe journal with meal planning, grocery scaling, and PDF export; integrated NVIDIA Nemotron through Nebius Token Factory for context-aware cooking assistance.

TECHNICAL SKILLS

Programming: Python, SQL, R, JavaScript | Frontend: React, Vite

AI and ML: PyTorch, Hugging Face Transformers, scikit-learn, LLMs, RAG, prompt engineering

Tools: LlamaIndex, ChromaDB, Ollama, Streamlit, Pandas, NumPy, Git; OpenAI and Claude APIs